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Department of Computer Science

Machine Learning

Perceptrons

Largest Irreducible Canonical Form

What arrangement of ones and zeros cannot be reduced further?

WX \ YZ				
	00	01	11	10
00				
01				
11				
10				

Largest Irreducible Canonical Form

What arrangement of ones and zeros cannot be reduced further?

WX \ YZ		00	01	11	10
00					
01					
11					
10					

Red=0
White=1

Largest Irreducible Canonical Form

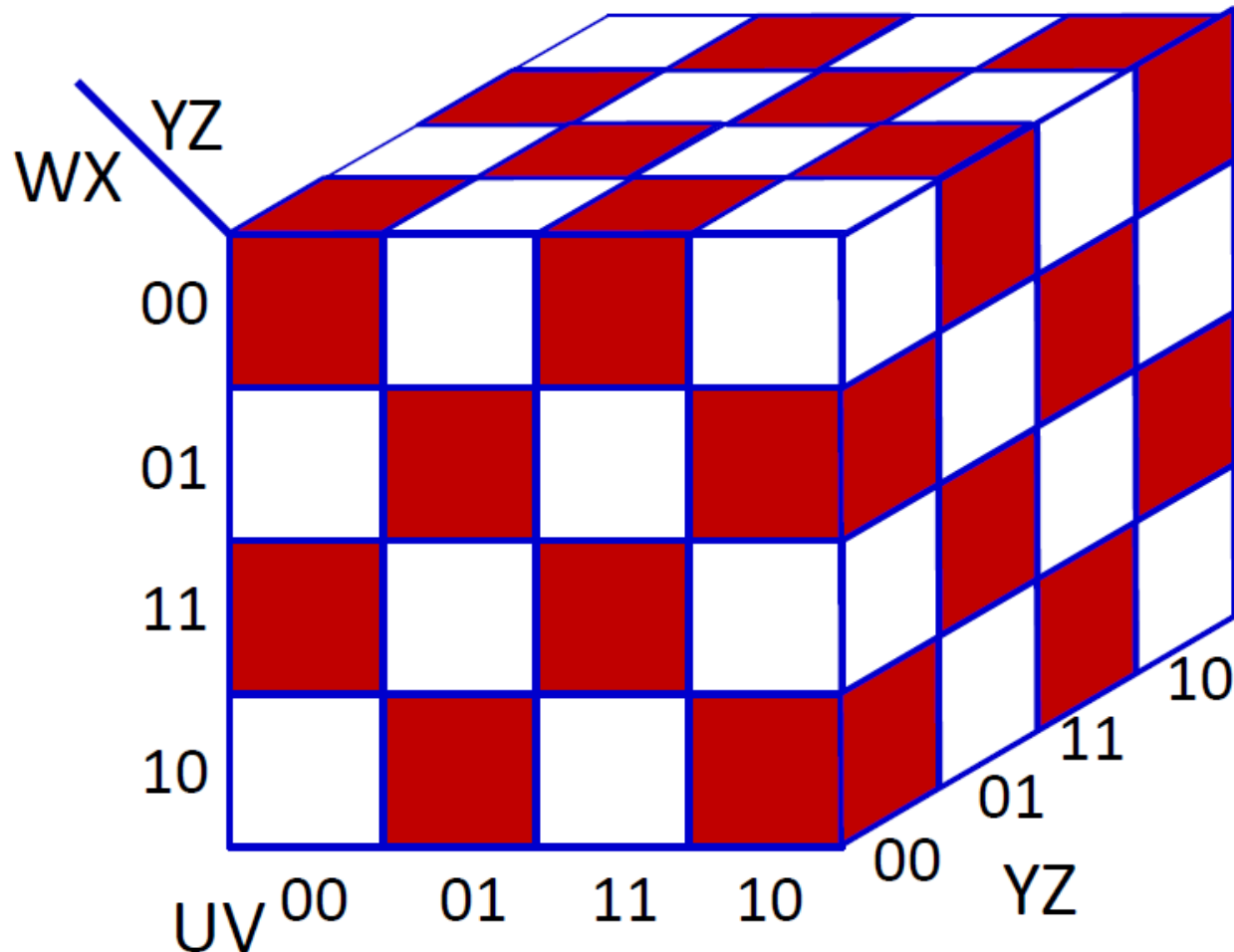
How many neurons for this Boolean function??

WX \ YZ	00	01	11	10
00	Red	White	Red	White
01	White	Red	White	Red
11	Red	White	Red	White
10	White	Red	White	Red

Red=0
White=1

Largest Irreducible Canonical Form

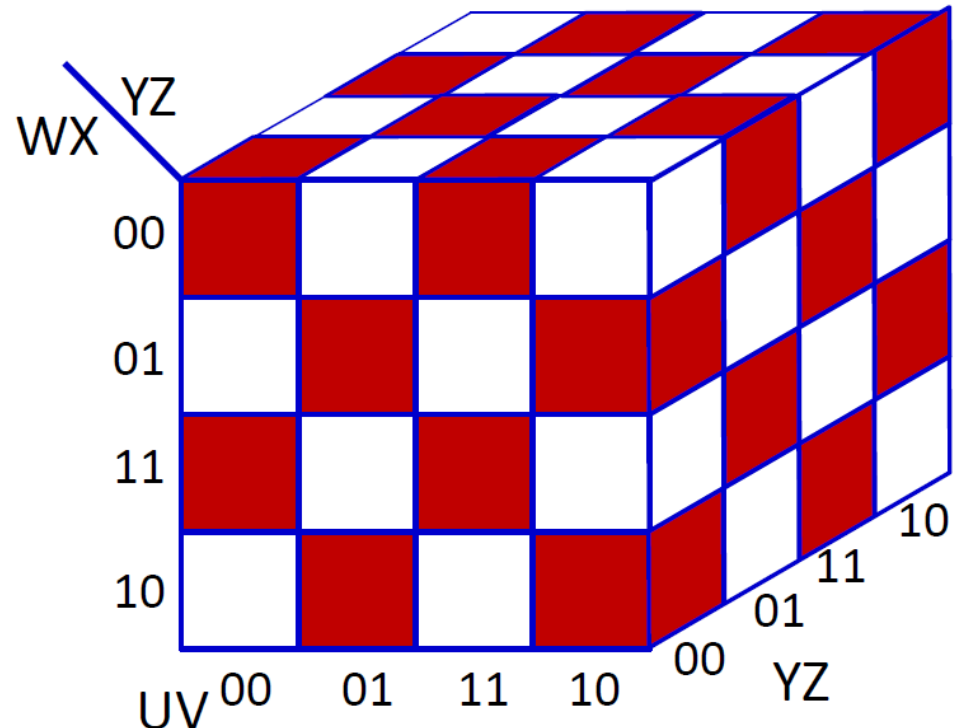
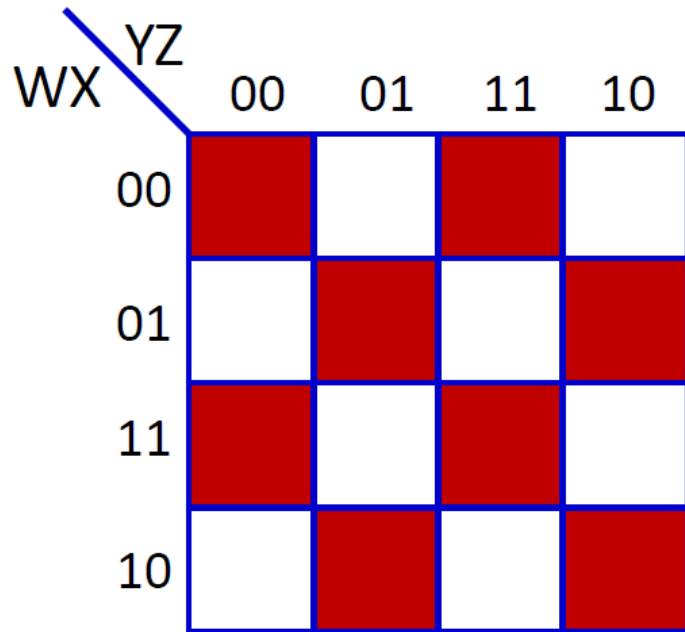
What about this function with 6 variables?



Largest Irreducible Canonical Form

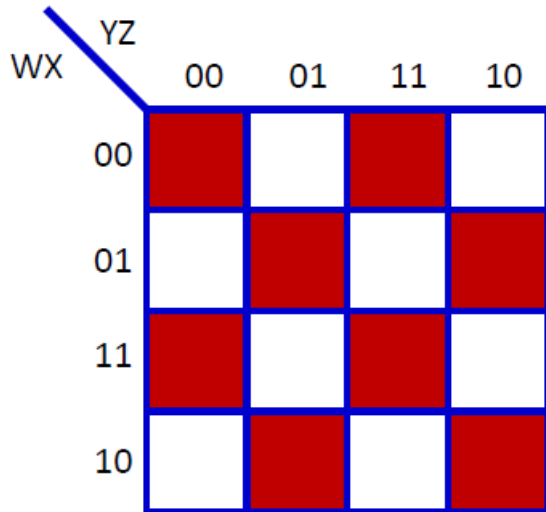
Can be generalized: Will require 2^{N-1} perceptrons in hidden layer

Exponential in N.

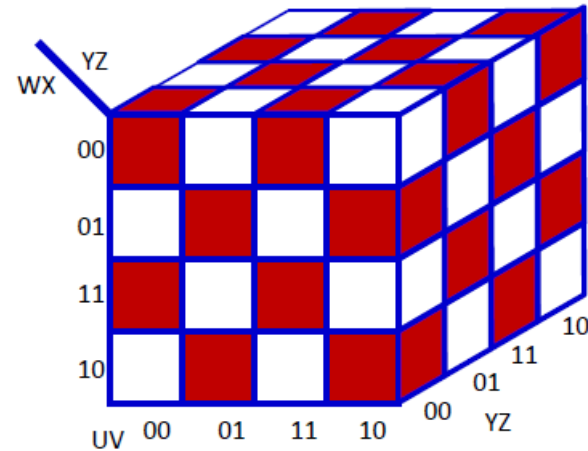


Which Boolean Function?

XOR!



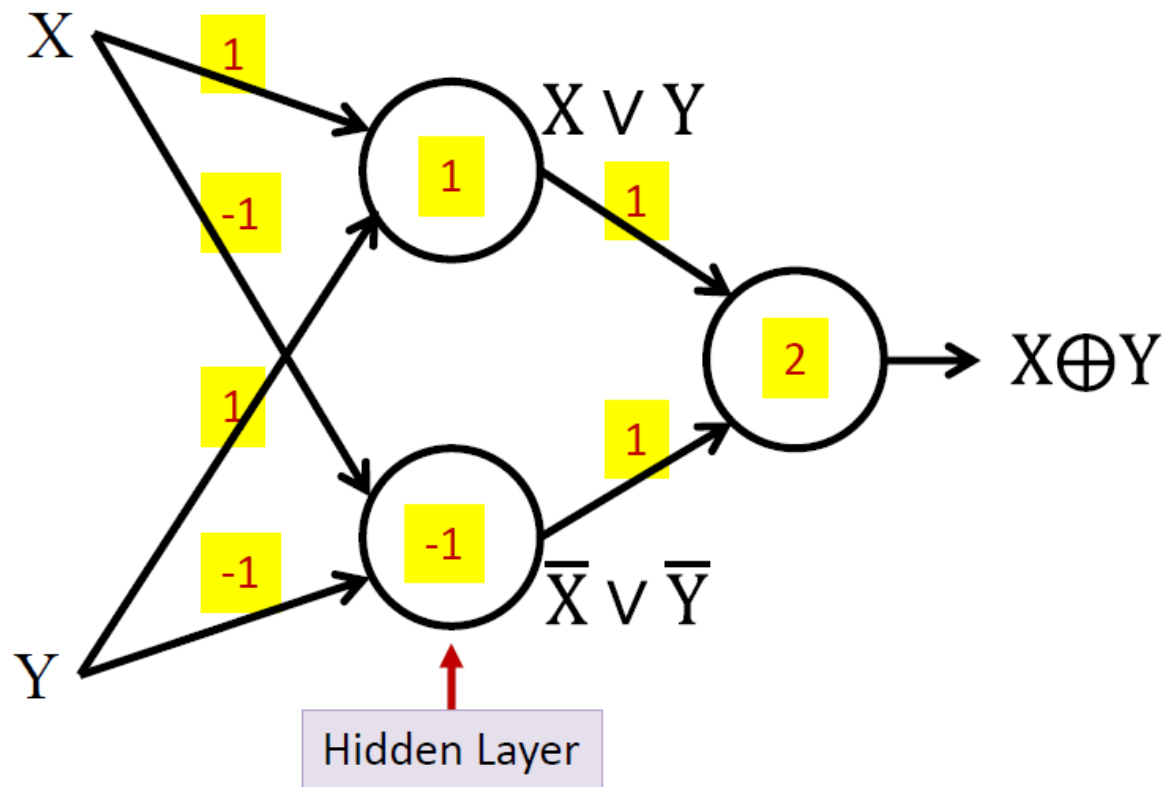
$$O = W \oplus X \oplus Y \oplus Z$$



$$O = U \oplus V \oplus W \oplus X \oplus Y \oplus Z$$

Which Boolean Function?

XOR!



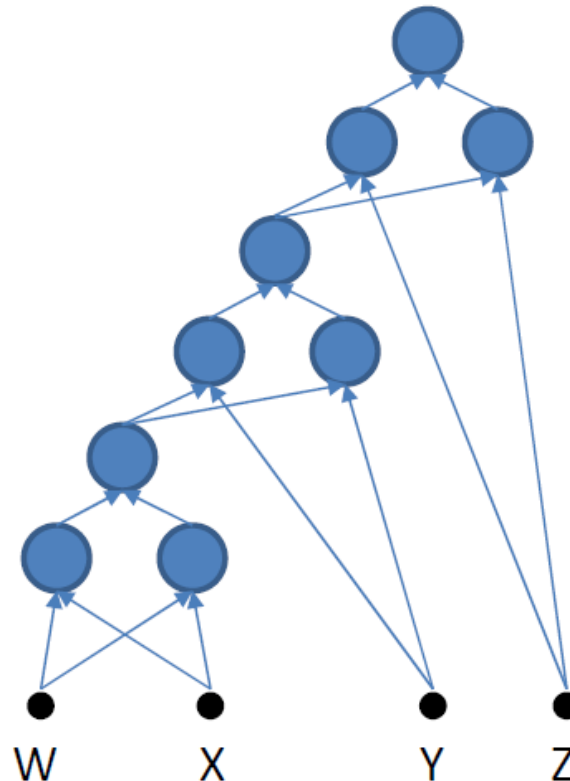
One Layer vs Multi-Layer Network

One XOR requires 3 perceptrons.

So, this network will require 9 perceptrons.

YZ \ WX	00	01	11	10
00	1	0	1	0
01	0	1	0	1
11	1	0	1	0
10	0	1	0	1

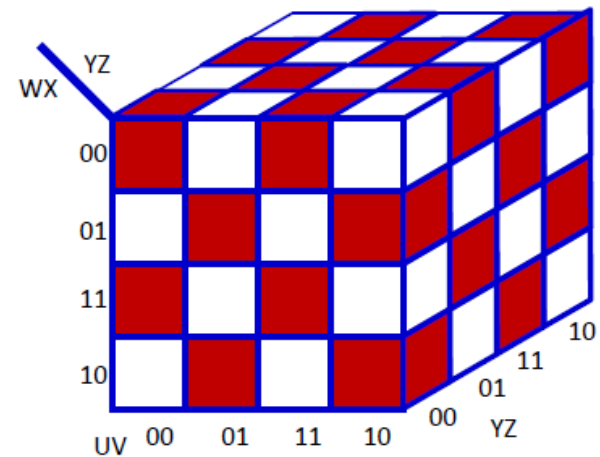
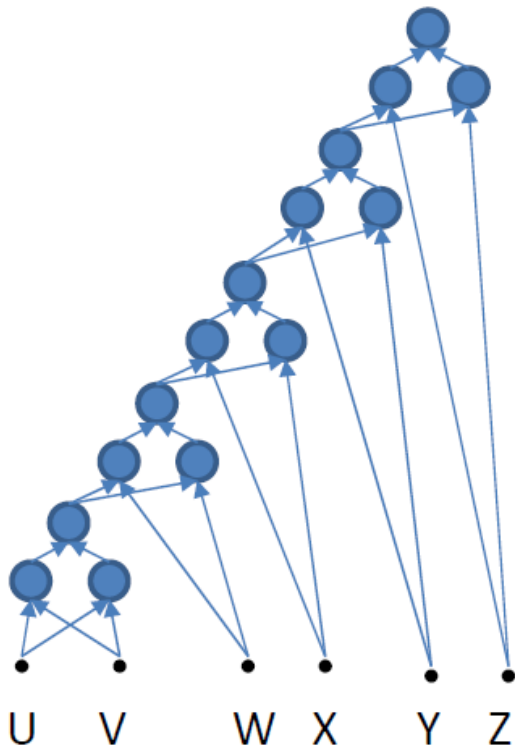
$$O = W \oplus X \oplus Y \oplus Z$$



9 perceptrons

One Layer vs Multi-Layer Network

Similarly, this network will require 15 perceptrons.



$$O = U \oplus V \oplus W \oplus X \oplus Y \oplus Z$$

15 perceptrons

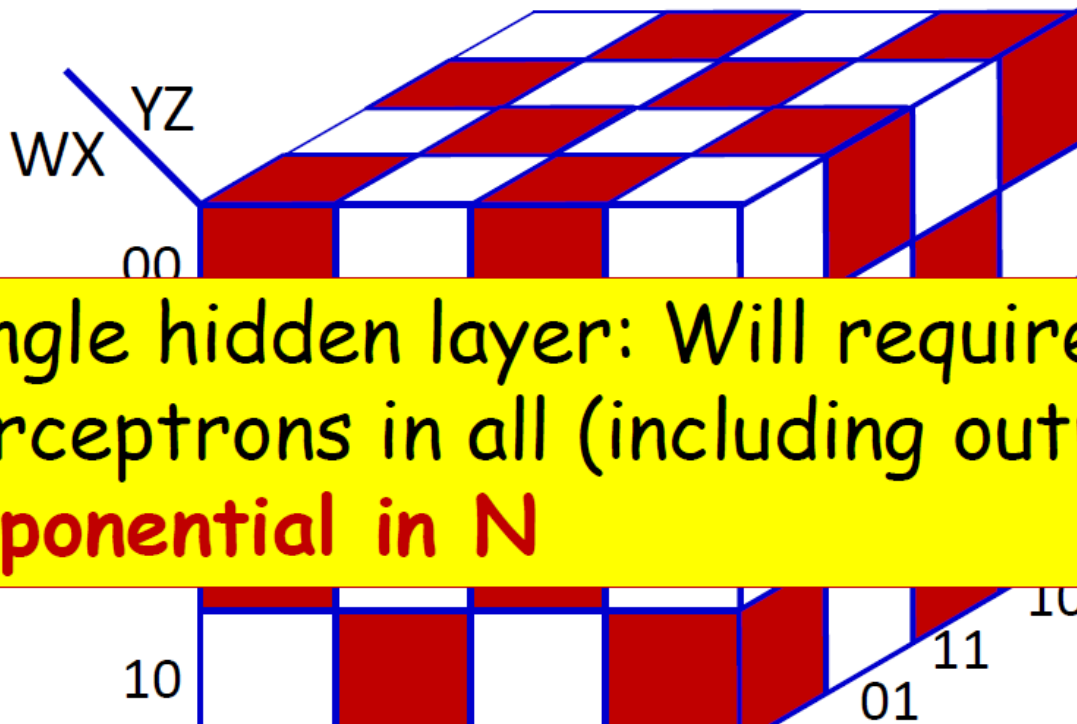
One Layer vs Multi-Layer Network

To generalize:

For N variables, the total number of perceptrons will be:

$$P_N = 3(N-1)$$

One Layer vs Multi-Layer Network



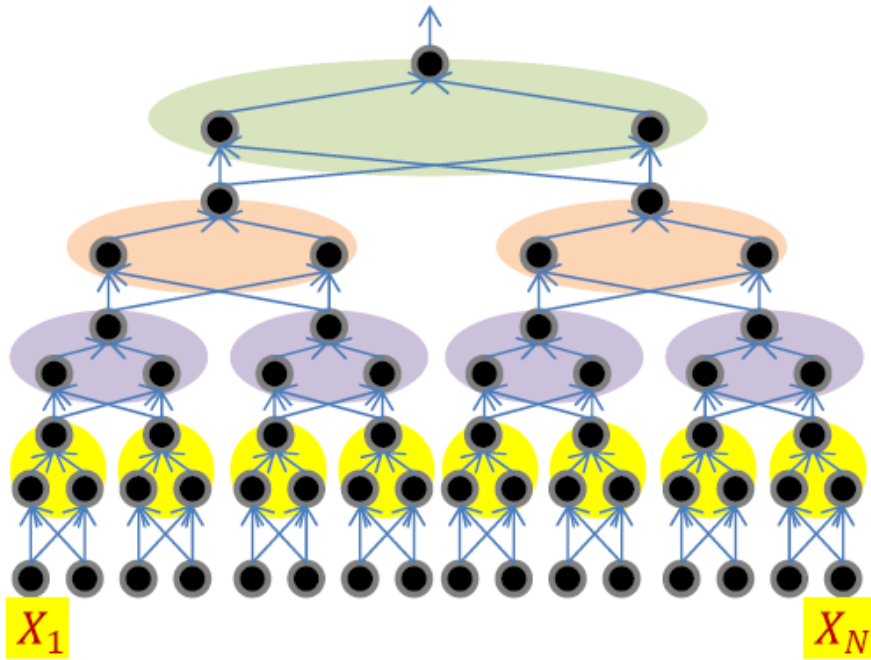
- Single hidden layer: Will require $2^{N-1}+1$ perceptrons in all (including output unit)
Exponential in N

Will require $3(N-1)$ perceptrons in a deep network

Linear in N!!!

Can be arranged in only $2\log_2(N)$ layers

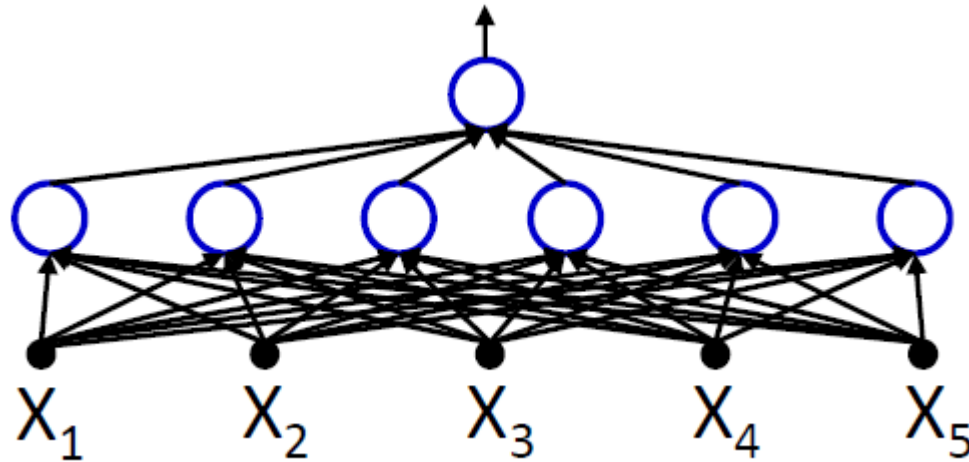
A better Design



$$O = X_1 \oplus X_2 \oplus \cdots \oplus X_N$$

Only $2\log_2 N$ layers.

Actual number of Parameters

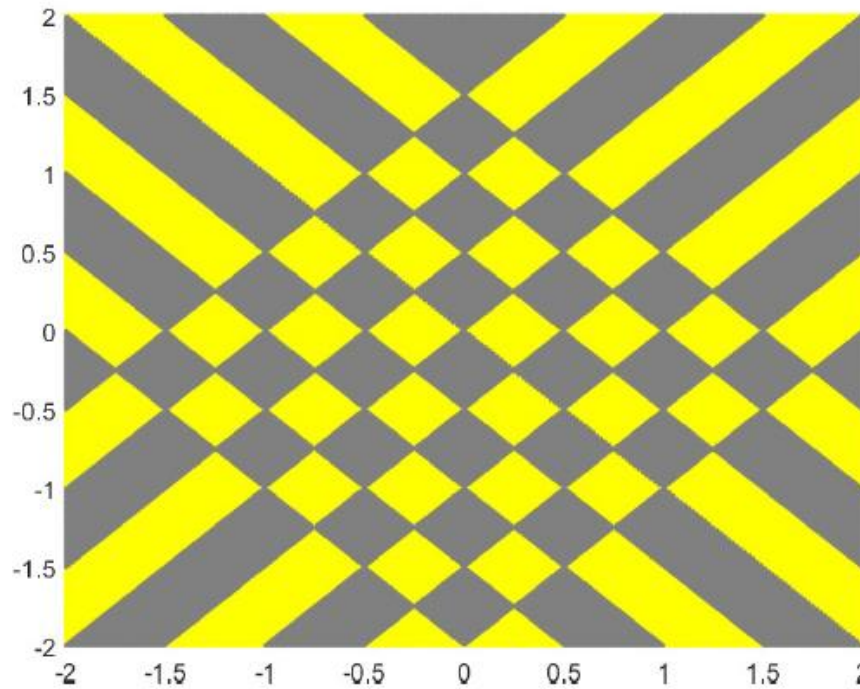


The actual number of parameters in a network is the number of connections

- In this example there are $30+5$

Networks that require an exponential number of neurons will require an exponential number of weights.

Design an MLP network

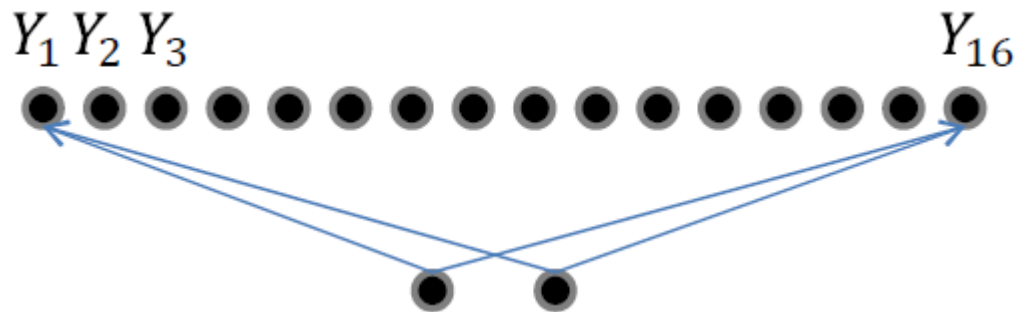
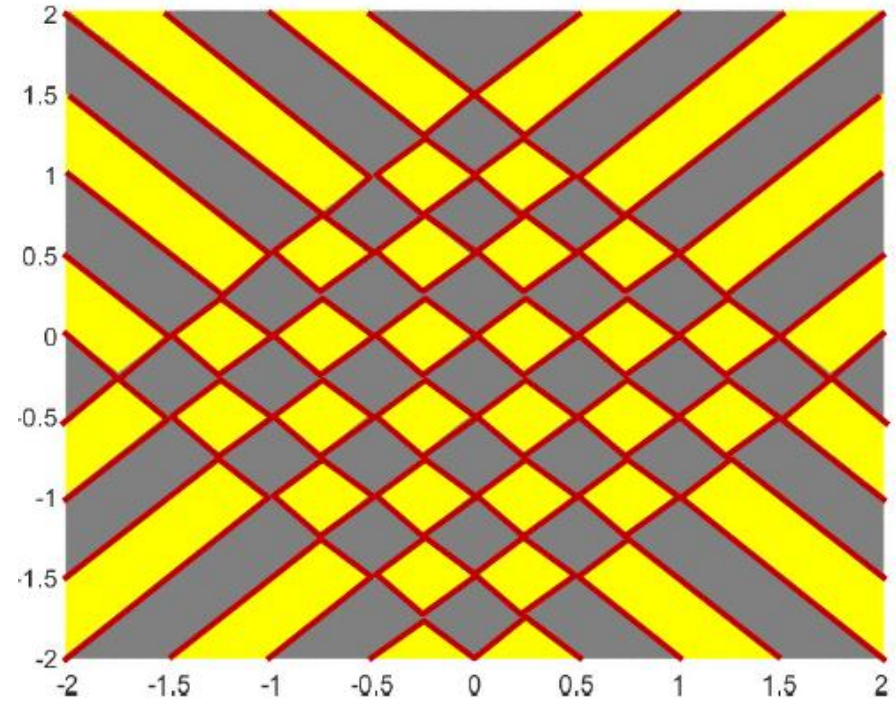


How many hidden layers will be required?

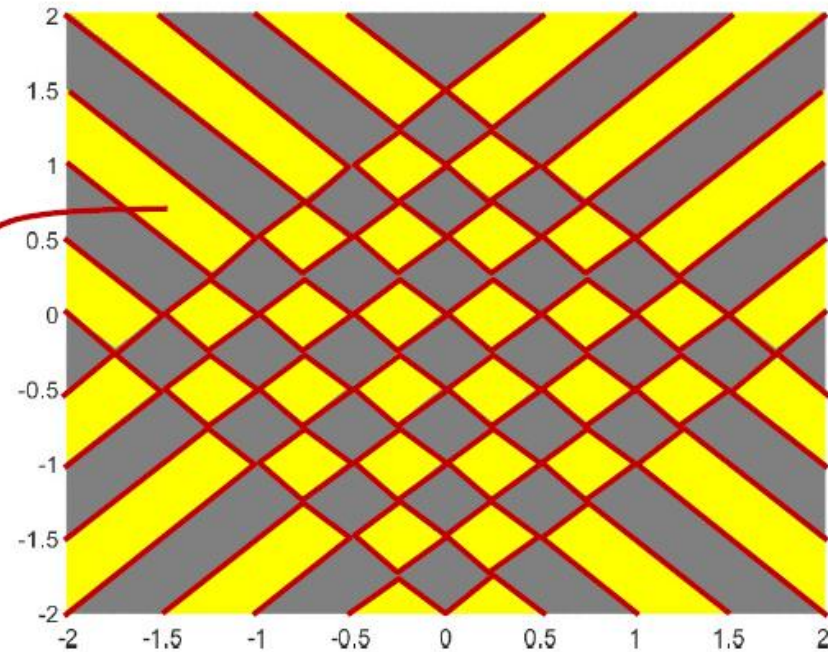
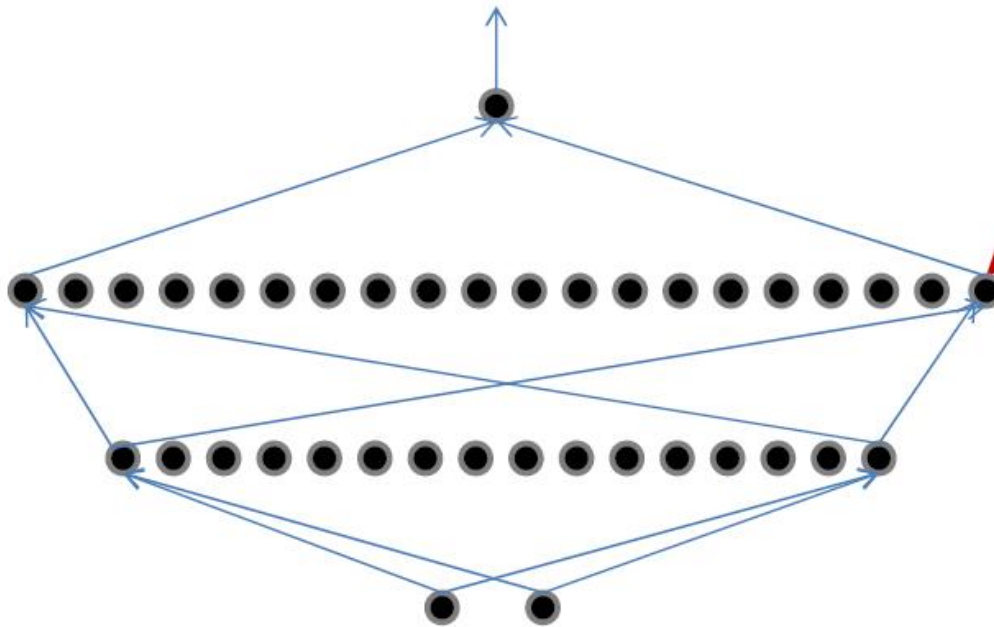
How many perceptrons at individual levels?

Design an MLP network

First layer will have 16 perceptron with these red classification lines.

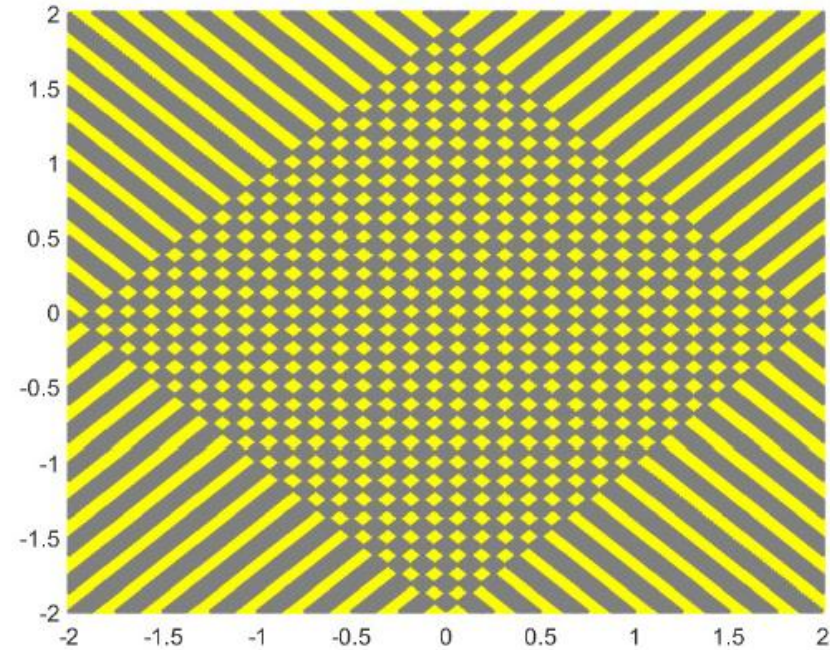
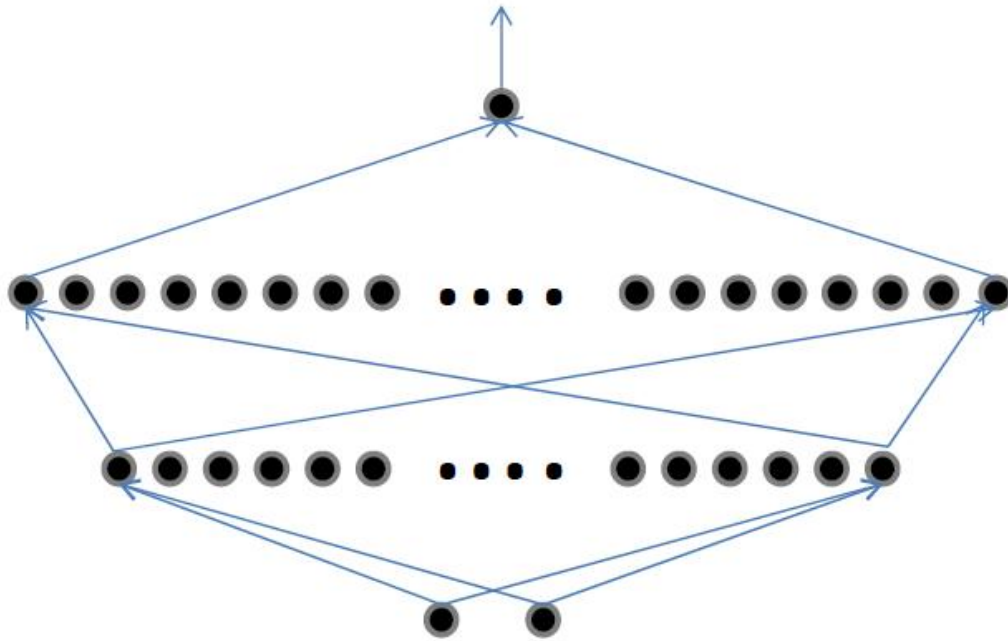


Design an MLP network



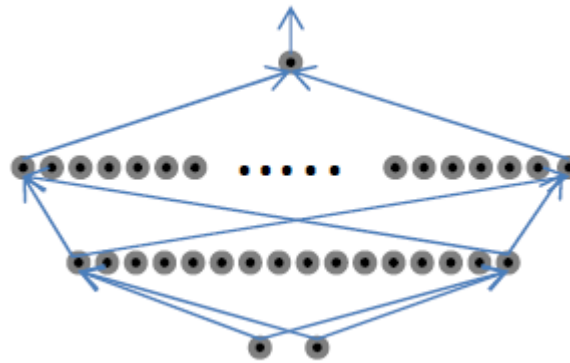
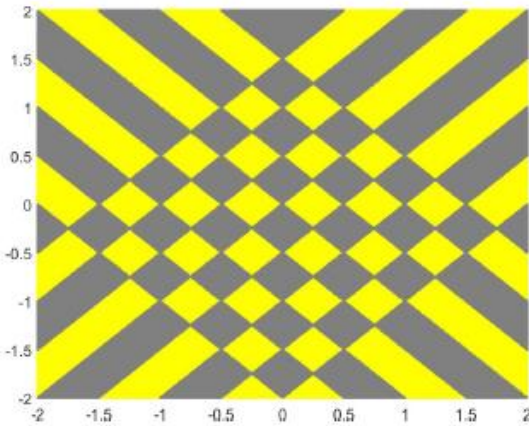
- Two-hidden-layer network: 56 hidden neurons
- 16 in hidden layer 1
 - 40 in hidden layer 2
 - 57 total neurons, including output neuron

Design an MLP network

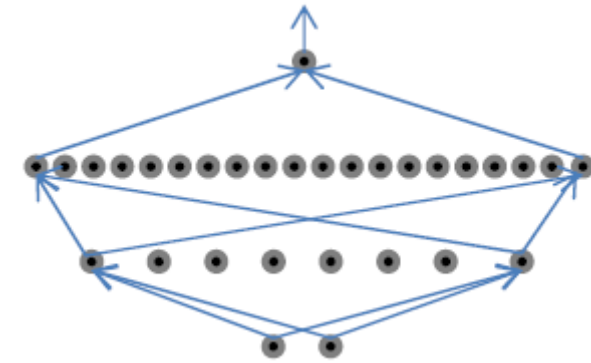


- Two-hidden-layer network: 608 hidden neurons
- 64 in hidden layer 1
 - 544 in hidden layer 2
 - 609 total neurons, including output neuron

Sufficiency of Architecture



A network with 16 or more perceptron in the first layer can represent the figure perfectly

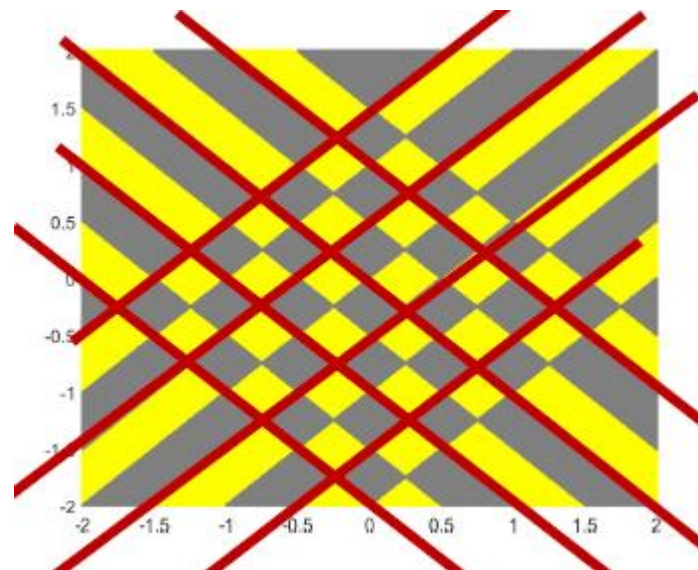


A network with less than 16 perceptrons in the first layer cannot represent this pattern exactly

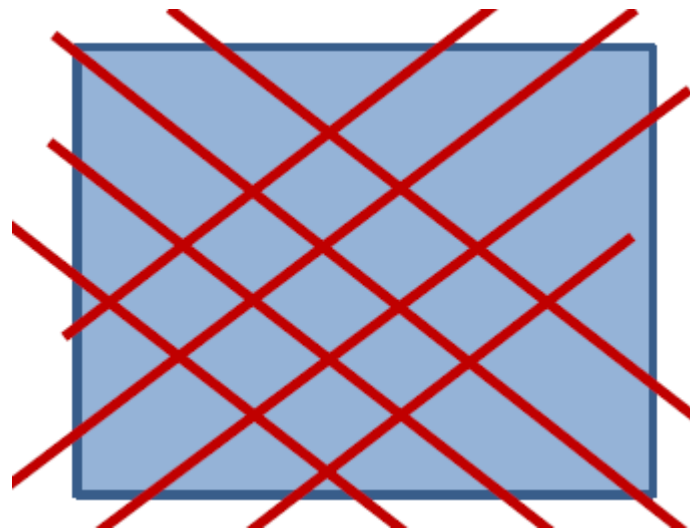
Not all designs can represent any function...

Design an MLP network

Even if the 8 first-layer neurons capture these boundaries...

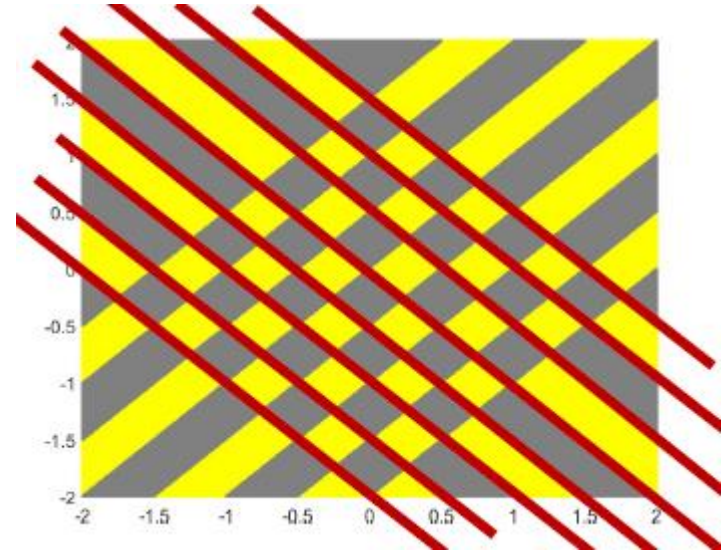


... they can only place you in one of these 25 cells but cannot inform you of where in the cell.

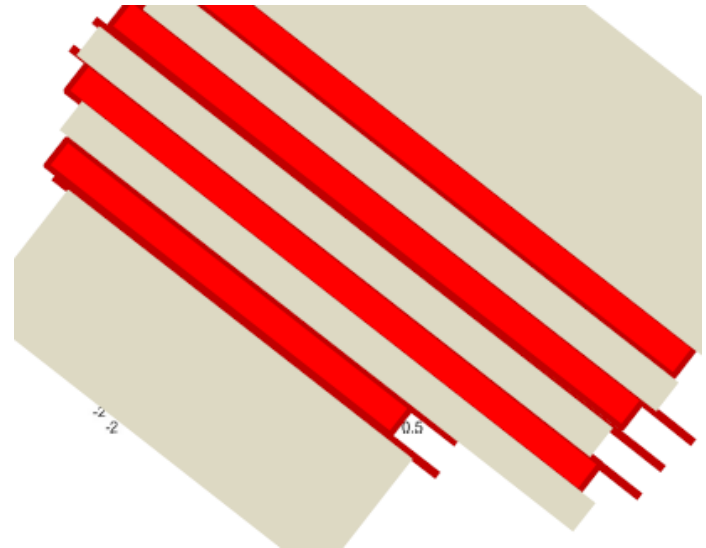


Design an MLP network

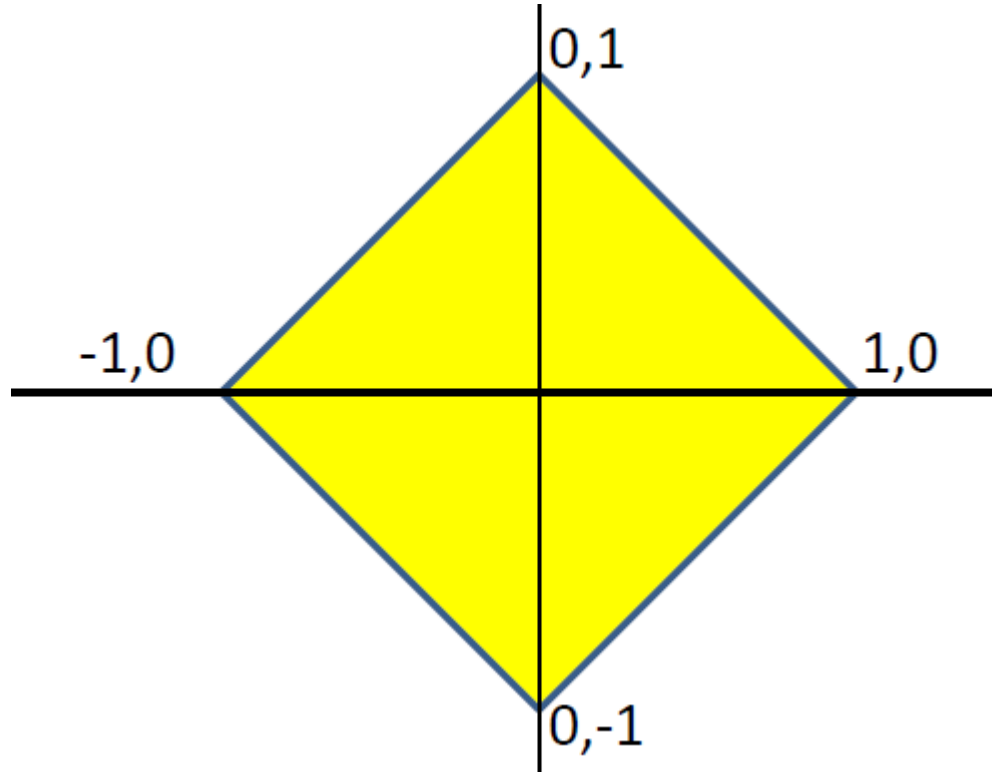
Similarly, if the 8 first-layer neurons capture these boundaries...



... they can only place you in one of these strips but not where in the strip.

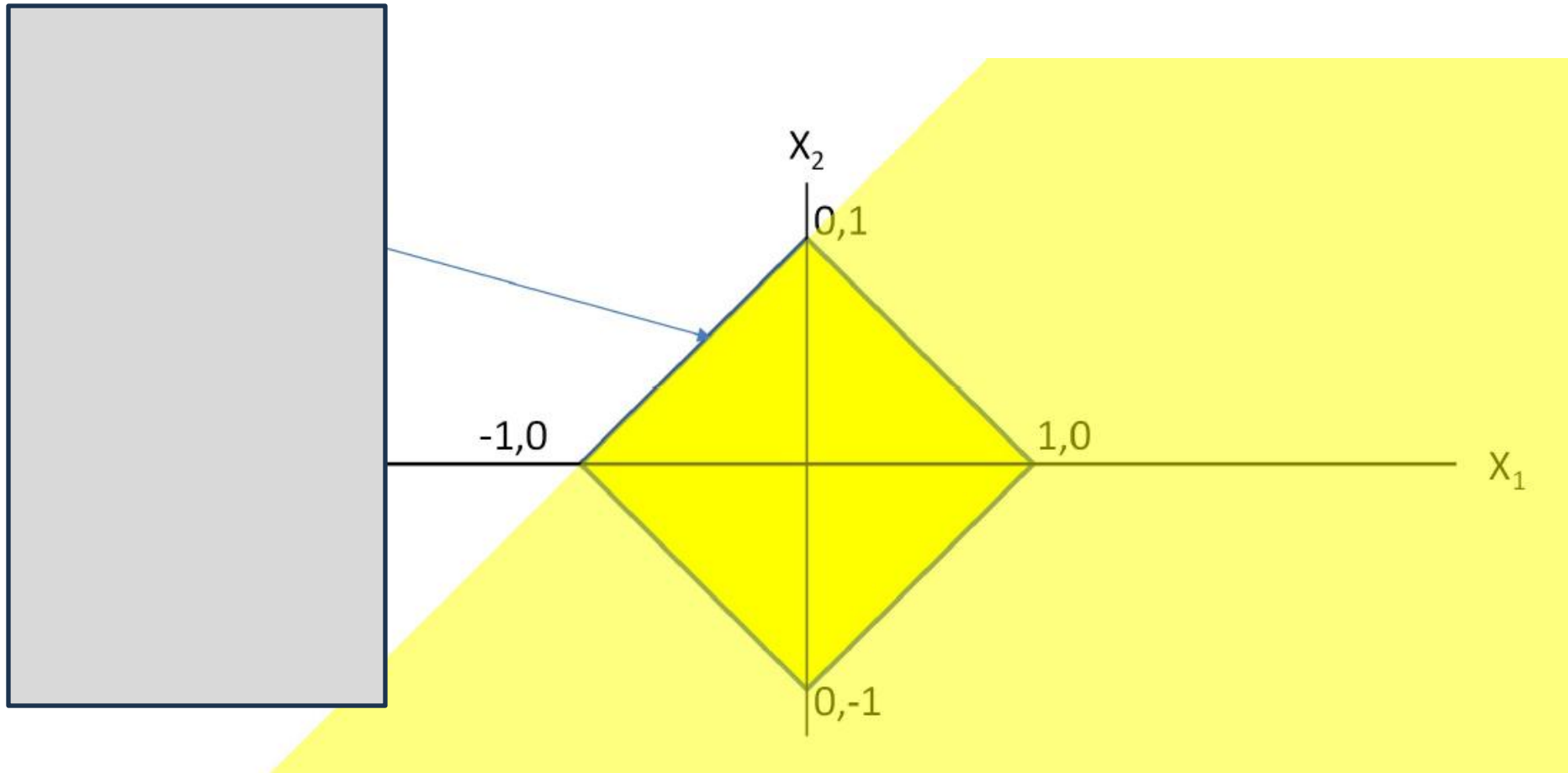


Construct a Network by hand

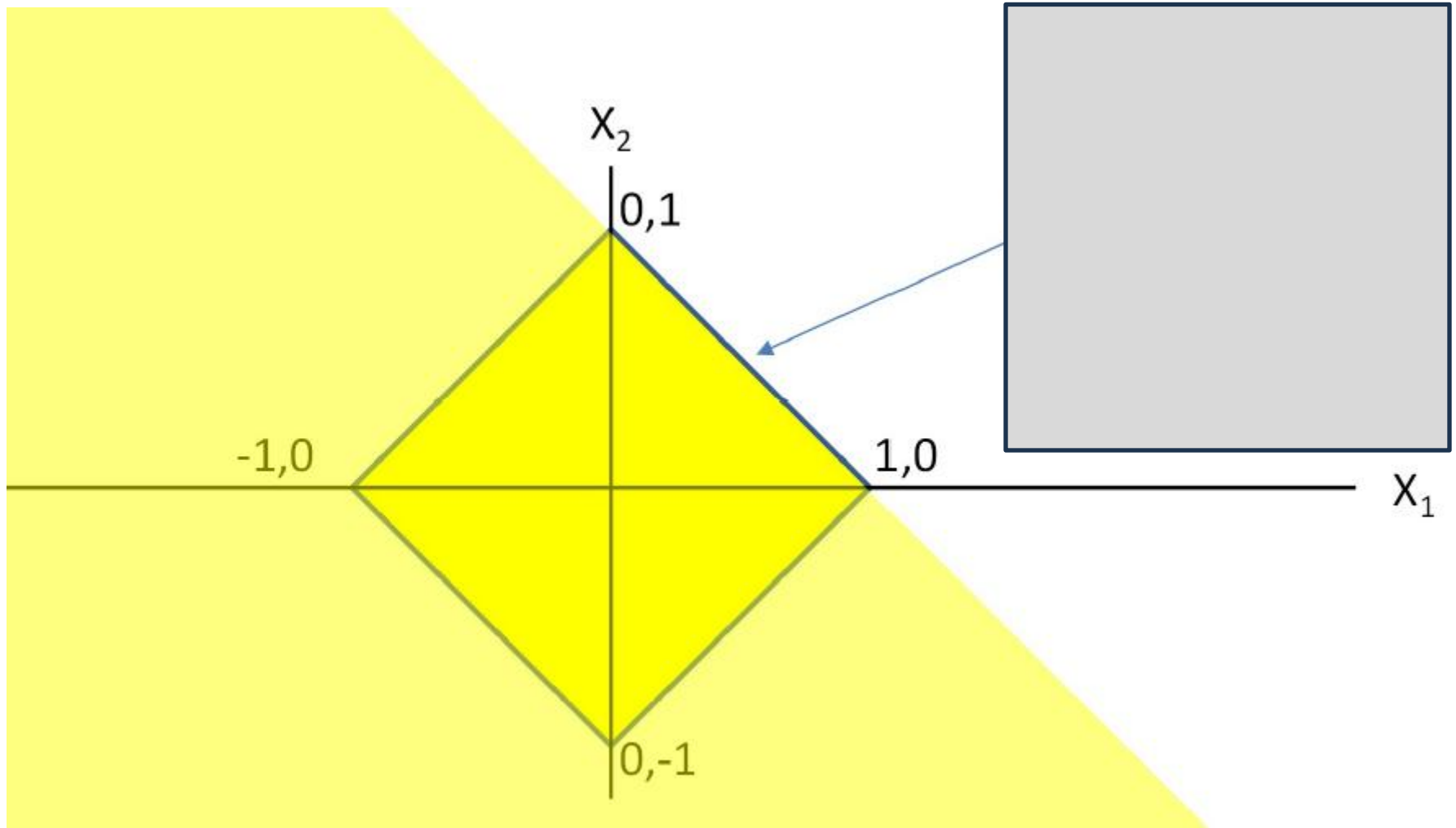


- Given a function, handcraft a network to satisfy it
- E.g.: Build an MLP to classify this decision boundary

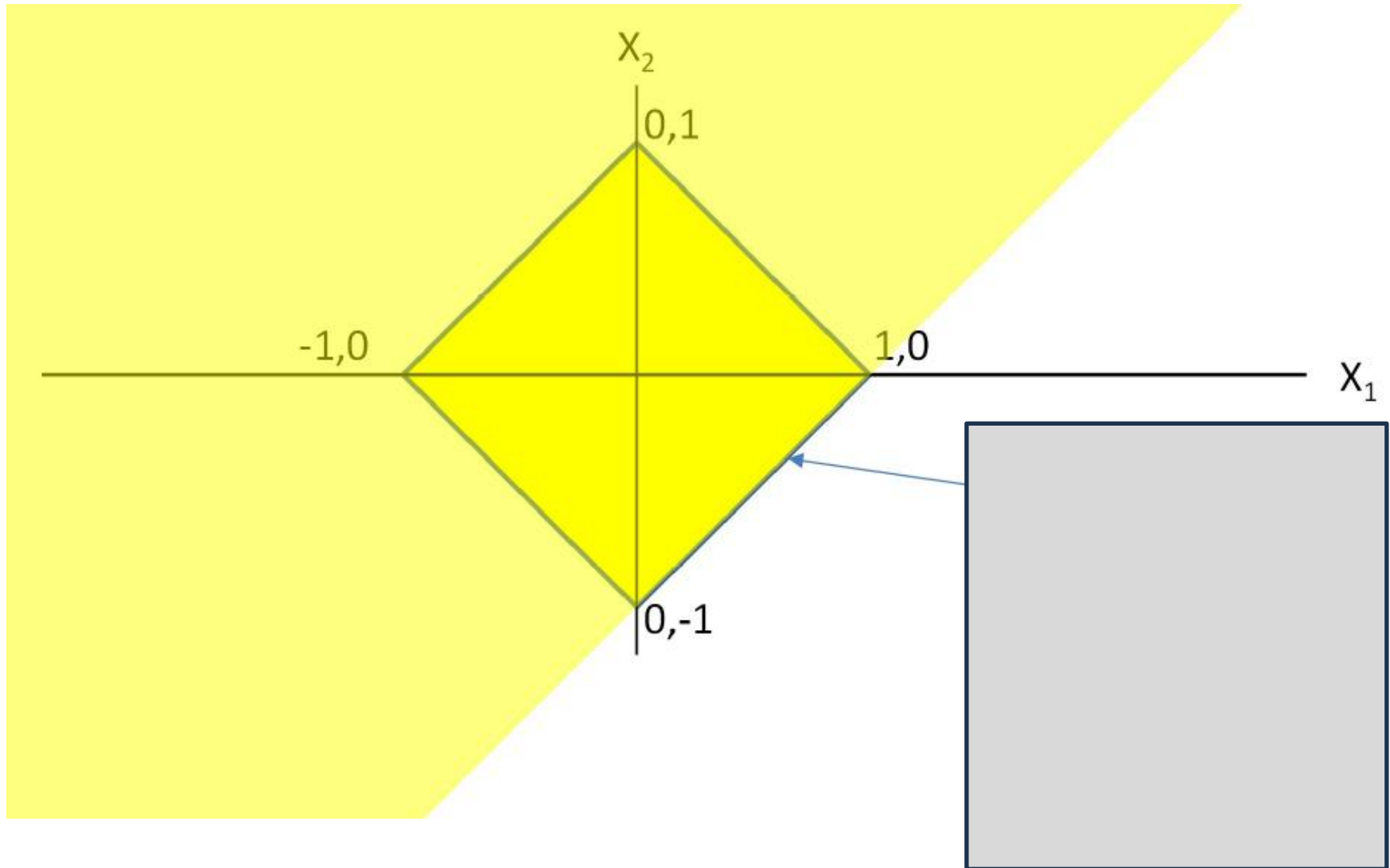
Construct a Network by hand



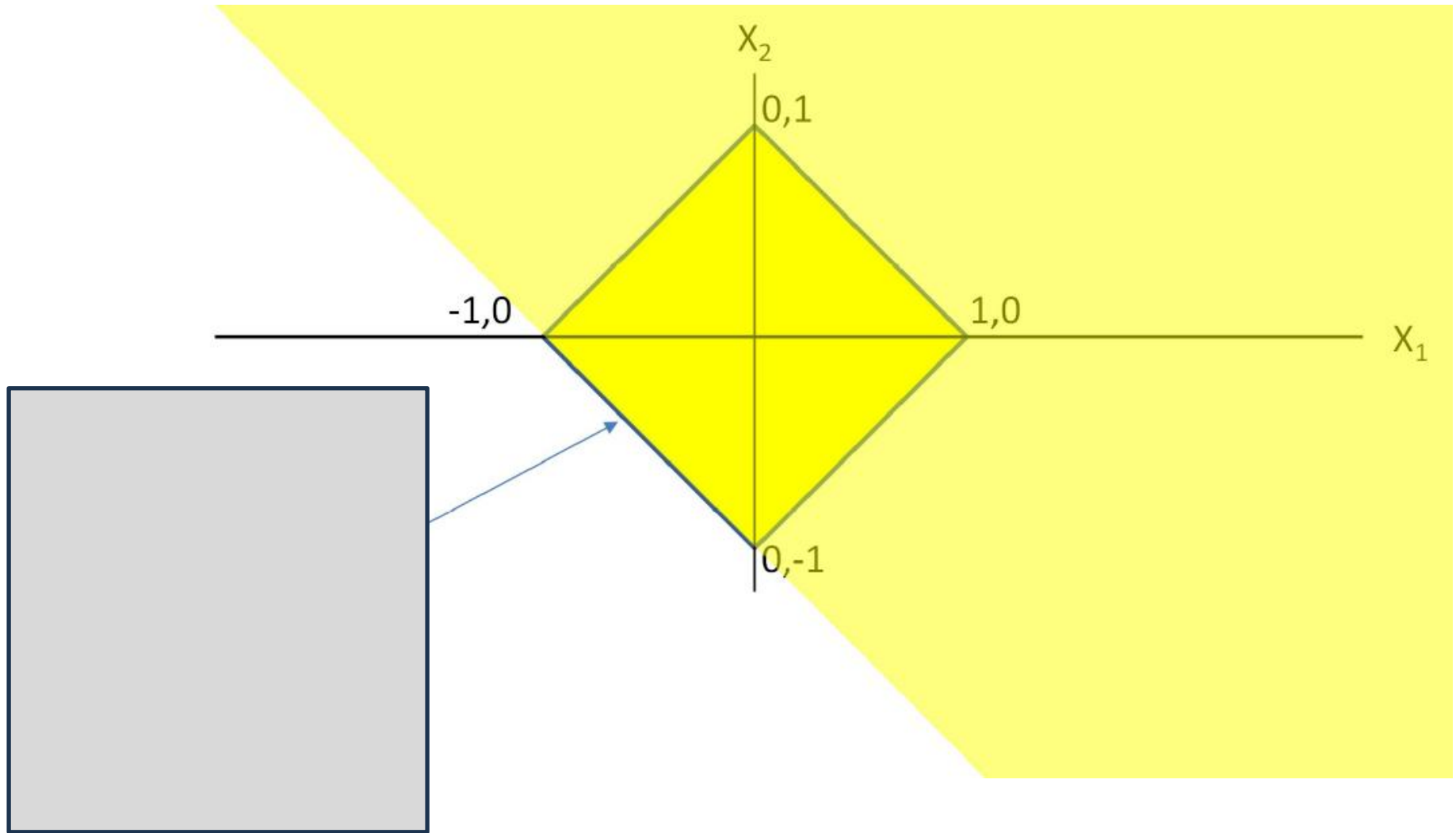
Construct a Network by hand



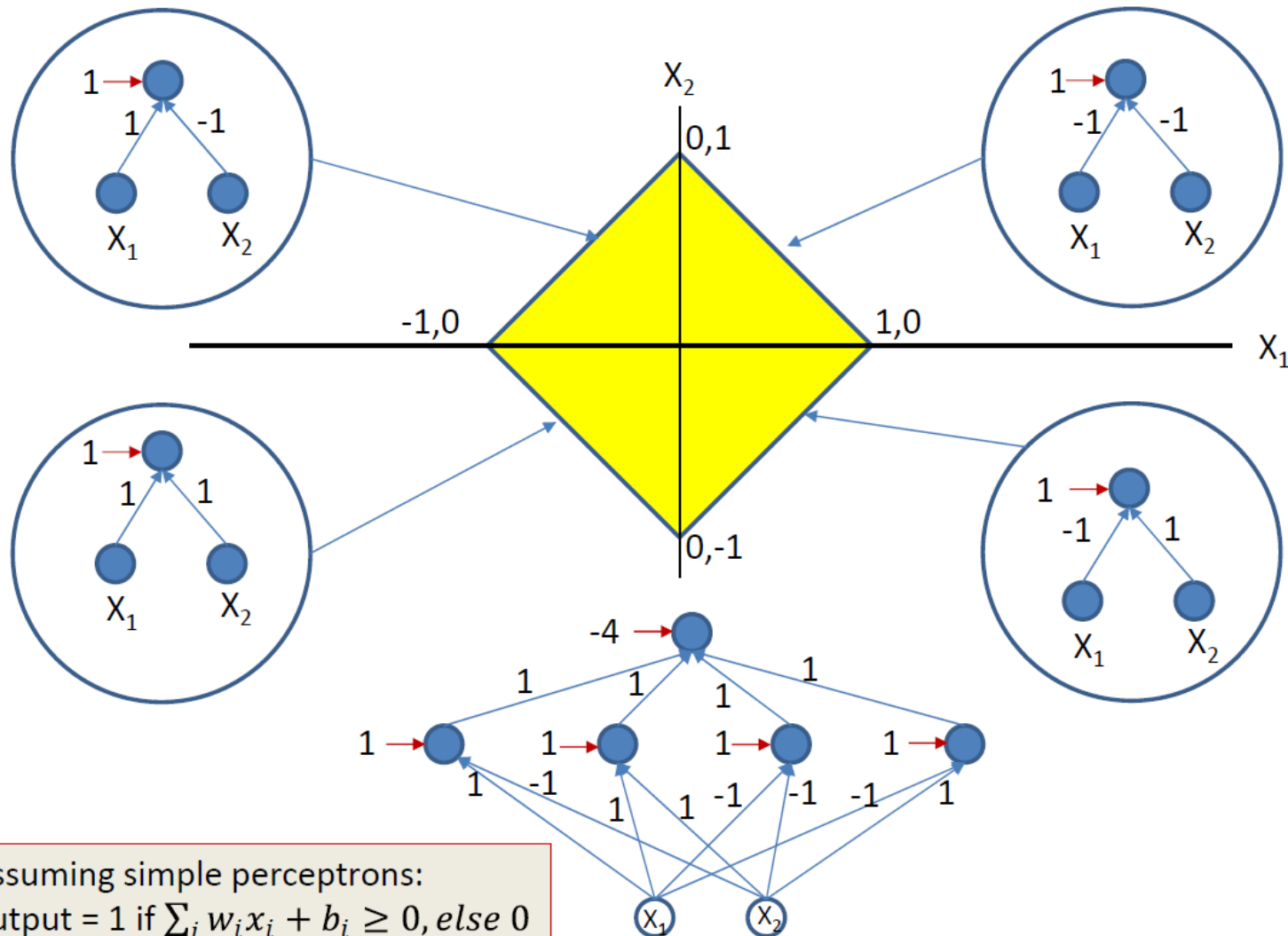
Construct a Network by hand



Construct a Network by hand



Construct a Network by hand



Thank you